

ETL Traffic Data Analysis Using PySpark

California Traffic Collision Data Analysis Report

Executive Summary

This report presents a comprehensive analysis of California traffic collision data using Apache Spark and PySpark. The analysis leverages the Statewide Integrated Traffic Records System (SWITRS) dataset to identify patterns in traffic collisions, assess risk factors, and provide data-driven recommendations for improving road safety.

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1. Introduction

Traffic collisions represent a significant threat to public safety, requiring continuous monitoring and analysis to develop effective preventive measures. This analysis examines California traffic collision data to uncover patterns related to accident severity, geographic distribution, temporal trends, and contributing environmental factors. By leveraging Apache Spark's distributed computing capabilities and AWS S3's scalable storage infrastructure, this project demonstrates how big data analytics can support evidence-based policy making and urban planning decisions.

2. Objectives and Business Value

2.1 Primary Objectives

The primary objectives of this analysis are to:

- Clean and prepare large-scale traffic collision data for analysis
- Identify patterns in collision severity, location, and timing
- Analyze environmental and infrastructural factors contributing to collisions
- Provide actionable insights for traffic safety improvements
- Store processed data in AWS S3 for efficient retrieval and scalability

2.2 Business Value

Traffic collisions pose significant risks to public safety and economic productivity. Government agencies, city planners, and policymakers require data-driven insights to:

- Improve infrastructure design and road safety measures
- Optimize traffic management systems
- Implement targeted preventive interventions
- Allocate resources effectively to high-risk areas
- Develop evidence-based safety policies

This analysis enables transportation authorities to process vast amounts of crash data efficiently, supporting faster and more informed decision-making that can ultimately reduce accident rates and save lives.

3. Dataset Overview

The dataset consists of California traffic collision data from the Statewide Integrated Traffic Records System (SWITRS). The data is structured into four primary tables:

3.1 Dataset Structure

4.2 Schema Inspection

Each DataFrame's schema was inspected using `printSchema()` to verify:

- Column names and data types
- Presence of required fields
- Data type consistency across tables

Key Findings:

- All tables loaded successfully with appropriate data types
 - Schema inference correctly identified numerical and categorical variables
 - No structural inconsistencies detected in the initial load
-

5. Data Cleaning

Data cleaning is critical for ensuring analysis accuracy and reliability. This section details the approaches used to handle missing values, fix column formatting issues, and address outliers.

5.1 Handling Missing Values

Approach:

Missing values can significantly impact analysis quality. A comprehensive approach was implemented to identify and handle both explicit NULL values and string-based null representations (e.g., "null", "na", "none", "").

Methodology:

A custom function `normalize_string_nulls()` was created to standardize all string-based null representations to proper NULL values across all DataFrames. This ensured consistent treatment of missing data throughout the analysis.

Code Implementation:

```
FAKE_NULLS = ["null", "na", "none", ""]
```

```
def normalize_string_nulls(df):
    for c, dtype in df.dtypes:
        df = df.withColumn(
            c,
            when(lower(trim(col(c))).isin(FAKE_NULLS), None)
            .otherwise(col(c))
        )
    return df
```

Missing Value Analysis:

The percentage of missing values was calculated for each column across all DataFrames. Only columns with missing values were displayed to focus attention on data quality issues.

Key Findings:

Collision DataFrame:

 Null Percentage Report for `collision\_df` (real + string NULLs)

column_name	null_percentage
caltrans_county	72.89
location_type	72.89
side_of_highway	72.89
latitude	71.5
longitude	71.5
caltrans_district	69.32
state_route	69.32
postmile	69.32
control_device	66.68
pcf_violation_subsection	64.27
reporting_district	59.06
chp_vehicle_type_at_fault	26.89
direction	24.86
statewide_vehicle_type_at_fault	19.89
beat_number	8.63
pcf_violation	6.86
special_condition	5.53
pcf_violation_category	1.65
intersection	1.01
road_surface	0.87

Victims DataFrame:

Null Percentage Report for `victims\_df` (real + string NULLs)

column_name	null_percentage
victim_safety_equipment_2	32.18
victim_safety_equipment_1	5.73
victim_age	3.33
victim_sex	2.49
victim_ejected	0.44
victim_seating_position	0.2

- After cleaning, victim_age was cast to integer type for proper numerical analysis

Parties DataFrame:

Null Percentage Report for `parties\_df` (real + string NULLs)

column_name	null_percentage
chp_vehicle_type_towed	48.72
party_safety_equipment_2	37.47
cellphone_in_use	30.44
party_race	23.4
cellphone_use_type	23.15
chp_vehicle_type_towing	21.49
party_safety_equipment_1	16.87
party_age	15.58
party_sex	13.35
statewide_vehicle_type	11.29
vehicle_make	10.45
vehicle_year	9.51
financial_responsibility	8.17
other_associate_factor_1	6.03
party_sobriety	2.98
direction_of_travel	2.81
movement_preceding_collision	1.78
party_type	0.24

Case IDs DataFrame:

- No null values in this dataframe

Decision: Rows with missing `victim_age` values were removed as age is critical for demographic analysis. For other columns, missing values were retained or handled based on their importance to specific analyses.

5.2 Fixing Columns

5.2.1 Data Type Corrections

Several columns required data type conversions to enable proper analysis:

- `victim_age`: Converted from string to integer type
- Timestamp columns: Converted to proper datetime format for temporal analysis

5.2.2 Duplicate Removal

A custom function was implemented to remove duplicate records across all DataFrames:

```
def remove_duplicates(df):  
    return df.dropDuplicates()
```

All four DataFrames were processed to eliminate duplicate records, ensuring data integrity.

5.3 Outlier Analysis

Methodology:

The Interquartile Range (IQR) method was employed to detect outliers in numerical columns, specifically `victim_age`.

IQR Outlier Detection Process:

1. Calculate Q1 (25th percentile) and Q3 (75th percentile)
2. Compute $IQR = Q3 - Q1$
3. Define outlier boundaries:
 - Lower bound = $Q1 - 1.5 \times IQR$
 - Upper bound = $Q3 + 1.5 \times IQR$
4. Flag values outside these boundaries as outliers

Implementation:

```
def detect_iqr_outliers_single_column(df, column_name, df_name):  
    quantiles = df_clean.approxQuantile(column_name, [0.25, 0.75], 0.01)
```

$q1, q3 = \text{quantiles}$

$iqr = q3 - q1$

$\text{lower} = q1 - 1.5 * iqr$

$\text{upper} = q3 + 1.5 * iqr$

Victim Age Outlier Analysis:

Results:

- Total rows analyzed: 931788
- Lower bound: -22.5
- Upper bound: 85.5
- Number of outliers detected: 4598
- Percentage of outliers: (0.49%)

Justification for Outliers:

Ages outside the range of 0-100 years were considered outliers and likely represent data entry errors or placeholder values (e.g., 999).

Treatment:

Outliers in victim_age were removed by filtering records to include only ages between 0 and 100:

```
victims_df = victims_df.filter(col("victim_age").between(0, 100))
```

Impact:

- Records before outlier removal: 9,63,933
 - Records after outlier removal: 9,31,655
-

6. Exploratory Data Analysis

6.1 Variable Classification

6.1.1 Categorical Variables

The following variables were classified as categorical and encoded using StringIndexer for analysis:

- officer_id, reporting_district, chp_shift, population
- county_location, beat_type, chp_beat_type, chp_beat_class
- beat_number, primary_road, secondary_road, direction
- weather_1, caltrans_county, location_type, side_of_highway
- collision_severity, primary_collision_factor, pcf_violation_category
- pcf_violation_subsection, hit_and_run, type_of_collision
- motor_vehicle_involved_with, pedestrian_action, road_surface
- road_condition_1, lighting, control_device
- statewide_vehicle_type_at_fault, chp_vehicle_type_at_fault

6.1.2 Numerical Variables

The following variables were classified as numerical:

- case_id, jurisdiction, county_city_location, special_condition
- distance, intersection, state_highway_indicator, caltrans_district
- state_route, postmile, tow_away, killed_victims, injured_victims
- party_count, pcf_violation, chp_road_type, pedestrian_collision
- bicycle_collision, motorcycle_collision, truck_collision
- not_private_property, severe_injury_count, other_visible_injury_count
- complaint_of_pain_injury_count, pedestrian_killed_count
- pedestrian_injured_count, bicyclist_killed_count, bicyclist_injured_count
- motorcyclist_killed_count, motorcyclist_injured_count
- latitude, longitude

6.1.3 String Indexing

Categorical variables were encoded using StringIndexer to convert string labels to numerical indices, enabling machine learning algorithms and statistical analysis:

```
indexers = [StringIndexer(inputCol=col, outputCol=f"{col}_idx",  
                          handleInvalid="keep")
```

```
for col in categorical_cols]
```

6.2 Distribution of Collision Severity

Objective: Analyze the distribution of collision severity levels to understand the proportion of minor versus severe incidents.

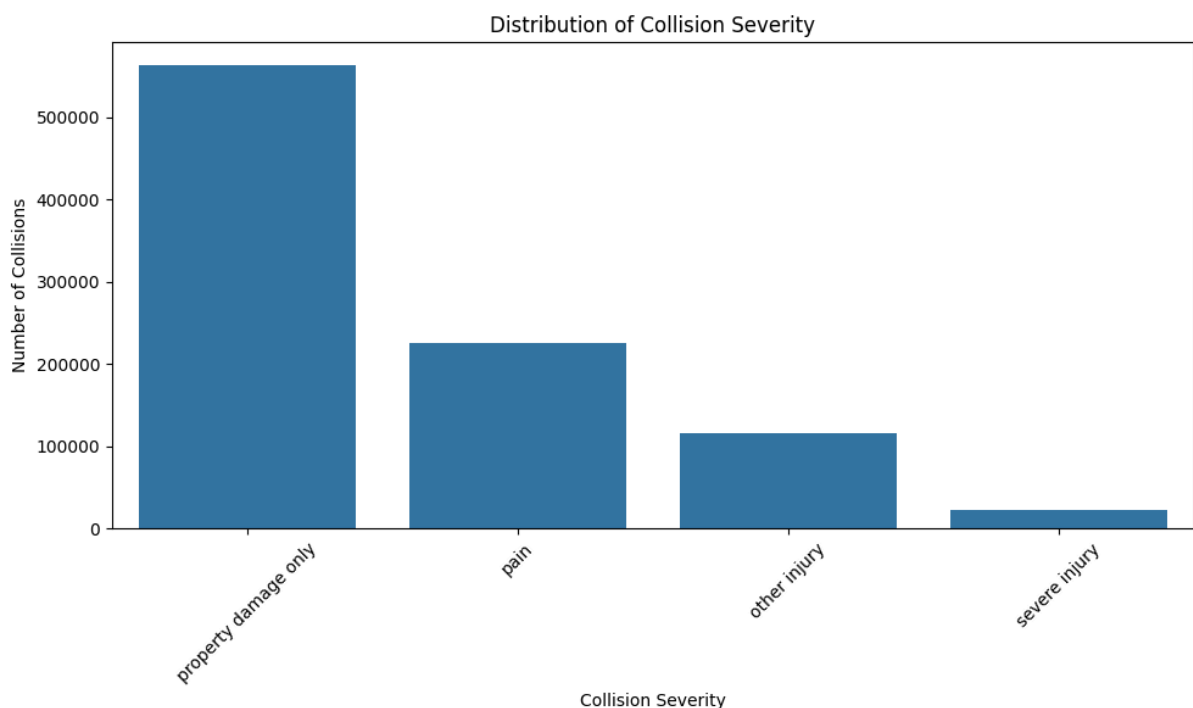
Methodology:

Collision severity was grouped and counted to create a frequency distribution. The data was visualized using a bar plot.

Key Findings:

Based on the bar chart visualization:

- Property Damage had the highest count for Collision severity followed by pain, other injury and severe injury.



Interpretation:

The distribution of collision severity provides critical insights into the nature of traffic incidents in California. Understanding severity patterns helps prioritize resource allocation and intervention strategies.

6.3 Weather Conditions During Collisions

Objective: Examine the distribution of weather conditions to identify which conditions are most prevalent during traffic collisions.

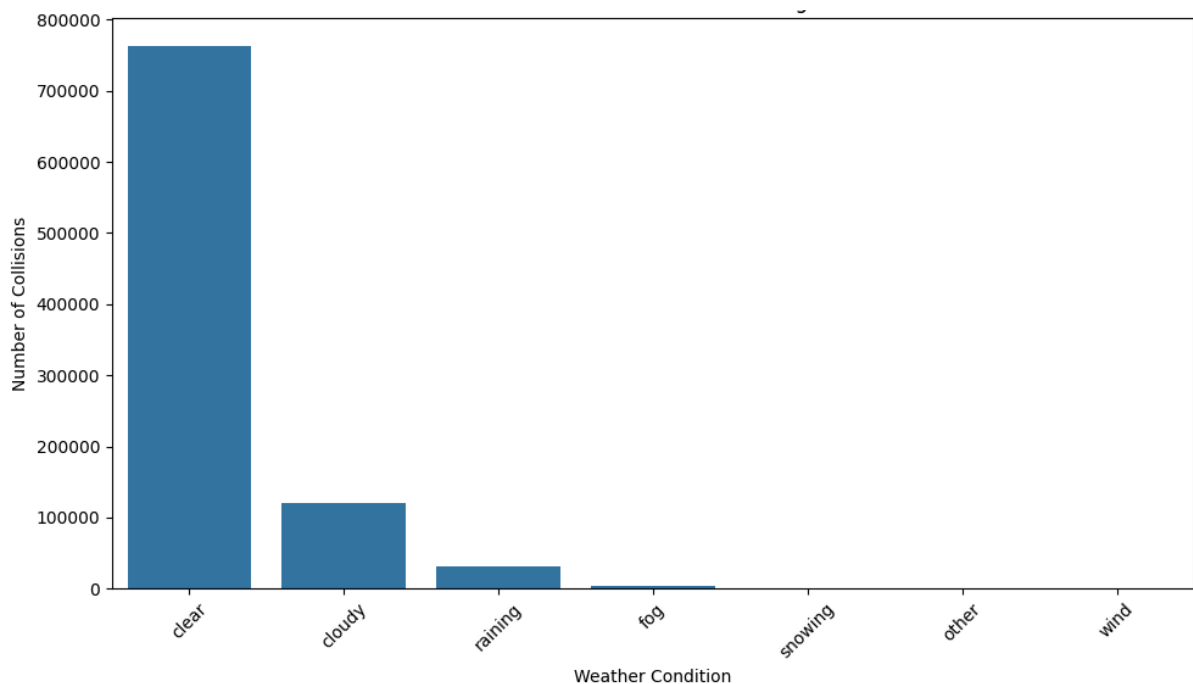
Methodology:

Weather conditions were grouped and counted, then sorted in descending order by frequency. Results were visualized using a horizontal bar chart.

Key Findings:

Based on the analysis:

- Most common weather condition: Clear
- Second most common: Cloudy



Interpretation:

The predominance of collisions in certain weather conditions may reflect both the frequency of those conditions and their impact on driving safety. Clear weather conditions showing high collision counts might indicate that most driving occurs in these conditions, rather than necessarily indicating higher risk.

6.4 Distribution of Victim Ages

Objective: Analyze the age distribution of collision victims to identify vulnerable age groups.

Methodology:

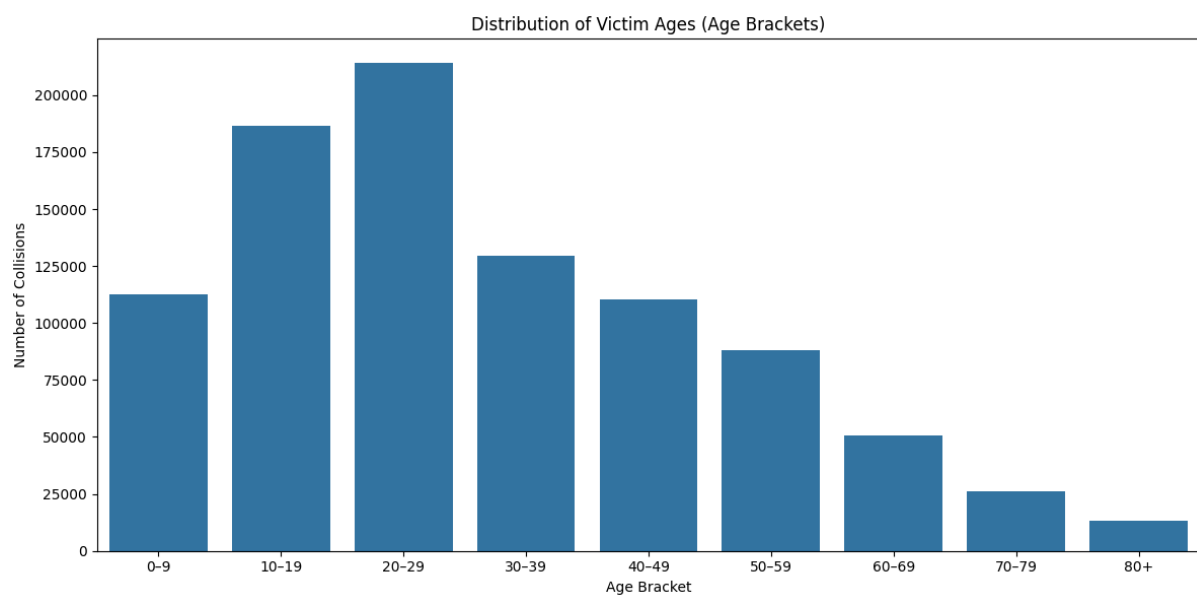
Two approaches were used:

1. **Individual Age Analysis:** Counted collisions for each specific age
2. **Age Bracket Analysis:** Grouped ages into brackets (0-9, 10-19, 20-29, 30-39, 40-49, 50-59, 60-69, 70-79, 80+) for clearer pattern identification

Key Findings:

Age Distribution:

- Peak age for collision victims: 18 years
- Age range with highest concentration: 20-29
- Least affected age bracket: 80+



Interpretation:

Age bracket analysis reveals that working-age adults (typically 20-49) are most frequently involved in traffic collisions, likely due to higher exposure through commuting and daily travel. Elderly populations (70+) show distinct patterns that may require targeted safety interventions.

6.5 Collision Severity vs Number of Victims

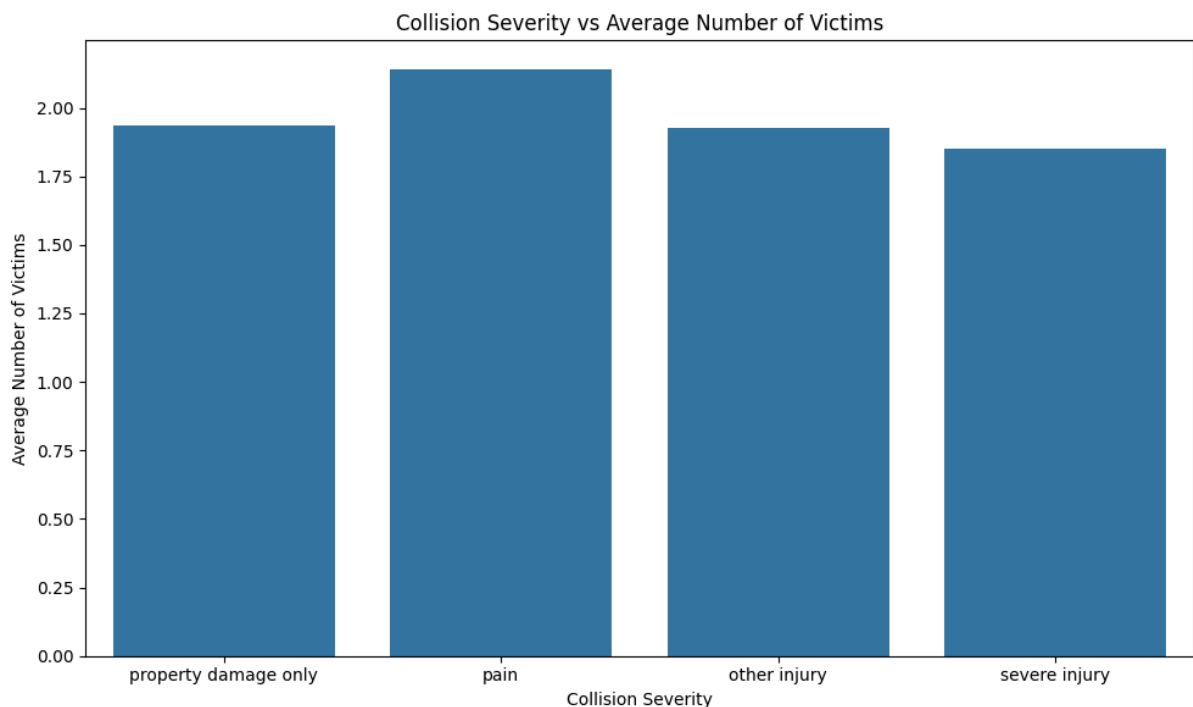
Objective: Study the relationship between collision severity and the average number of victims (party count) to understand if more severe collisions involve more parties.

Methodology:

The average party count was calculated for each collision severity level using bivariate analysis. Results were visualized using a bar chart.

Key Findings:

- Collision severity with highest average party count: Pain
- Average victims in fatal collisions: ~1.75
- Average victims in property damage only: >1.75



Interpretation:

The relationship between severity and party count reveals important patterns about collision dynamics. Multi-vehicle collisions may have different severity profiles compared to single-vehicle incidents, informing traffic management strategies.

6.6 Weather Conditions vs Collision Severity

Objective: Analyze the correlation between weather conditions and collision severity to identify high-risk weather scenarios.

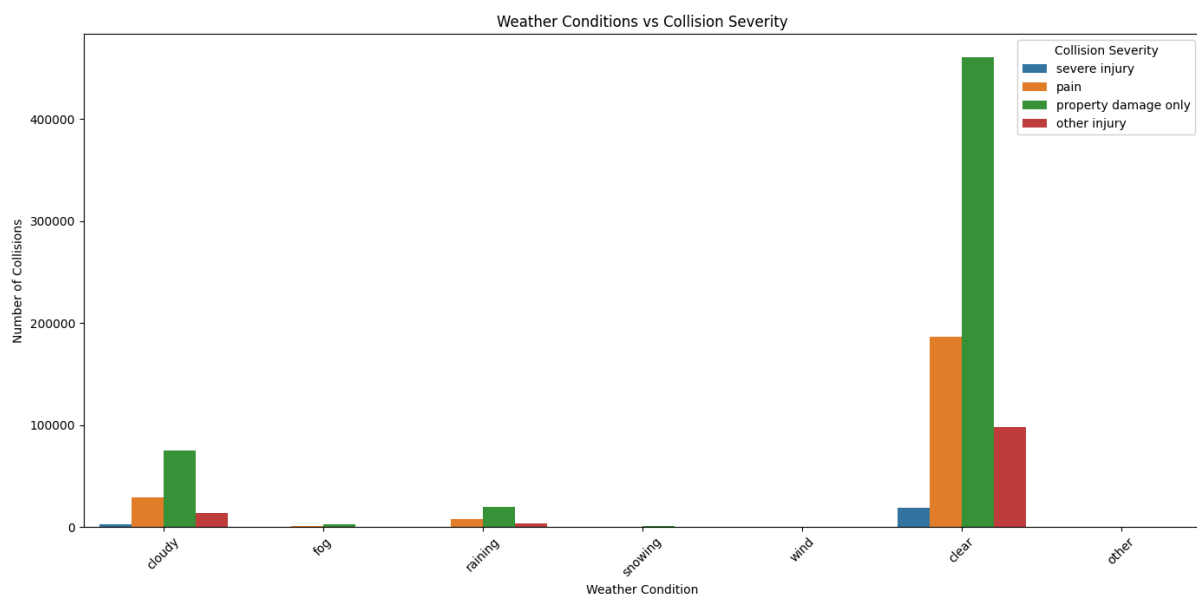
Methodology:

A grouped bar chart was created showing collision counts for each combination of weather condition and severity level. Additionally, severe injury ratios were calculated for each weather type.

Key Findings:

Severe Injury Ratio by Weather:

- Highest severe injury ratio: Wind (ratio: 0.0366)
- Second highest: Fog (ratio: 0.036)
- Weather conditions with disproportionately severe outcomes identified



Interpretation:

While clear weather may have the highest absolute collision count, wind and fog conditions show the highest severe injury ratios. This indicates that these conditions, though less frequent, result in disproportionately dangerous collisions. This finding suggests:

- Enhanced warnings needed for wind and fog conditions
- Potential for speed limit adjustments during adverse weather

- Infrastructure improvements in fog-prone areas

6.7 Lighting Conditions vs Collision Severity

Objective: Visualize the impact of lighting conditions on collision severity.

Methodology:

Collisions were grouped by lighting condition and severity level. A grouped bar chart was created, and severe injury ratios were calculated for each lighting condition.

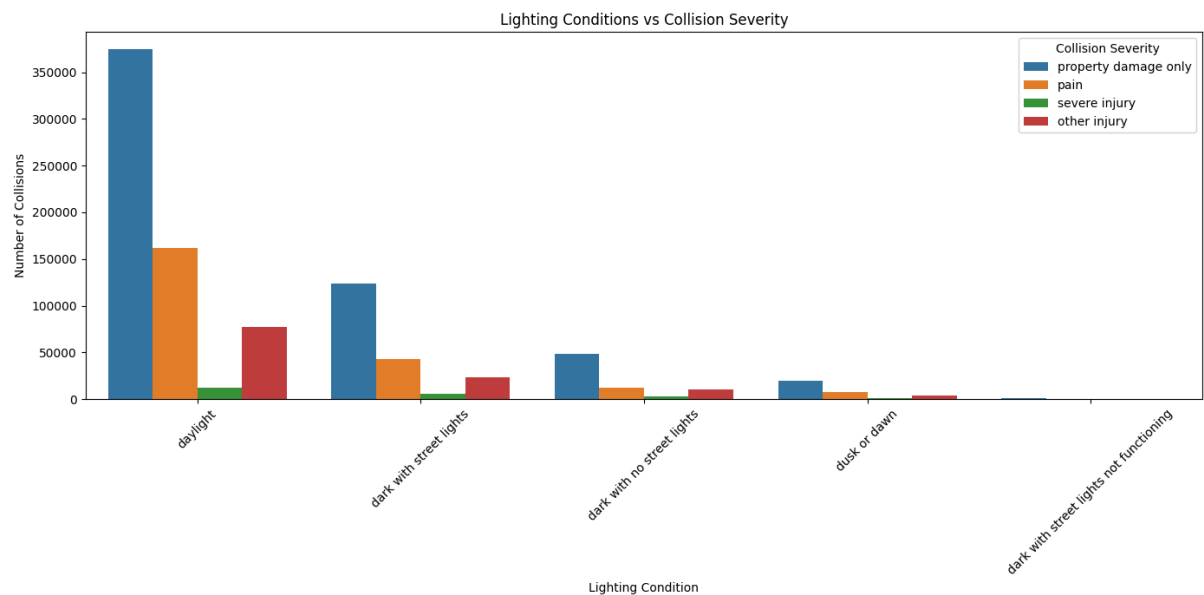
Key Findings:

Collision Distribution by Lighting:

- Most collisions occur in: Daylight
- Dark with no street lights shows highest severe injury ratio

Severe Injury Ratio Analysis:

- Dark with no street lights: 0.043 (highest ratio)
- Daylight: 0.019 (lowest in terms of ratio)



Interpretation:

The analysis reveals that "dark with no street lights" conditions have the highest severe injury ratio, indicating significantly increased risk in poorly lit areas. This finding has important policy implications:

- Prioritize street lighting installation in high-collision areas
- Enhanced nighttime enforcement in unlit areas
- Reflective road markings and signage improvements

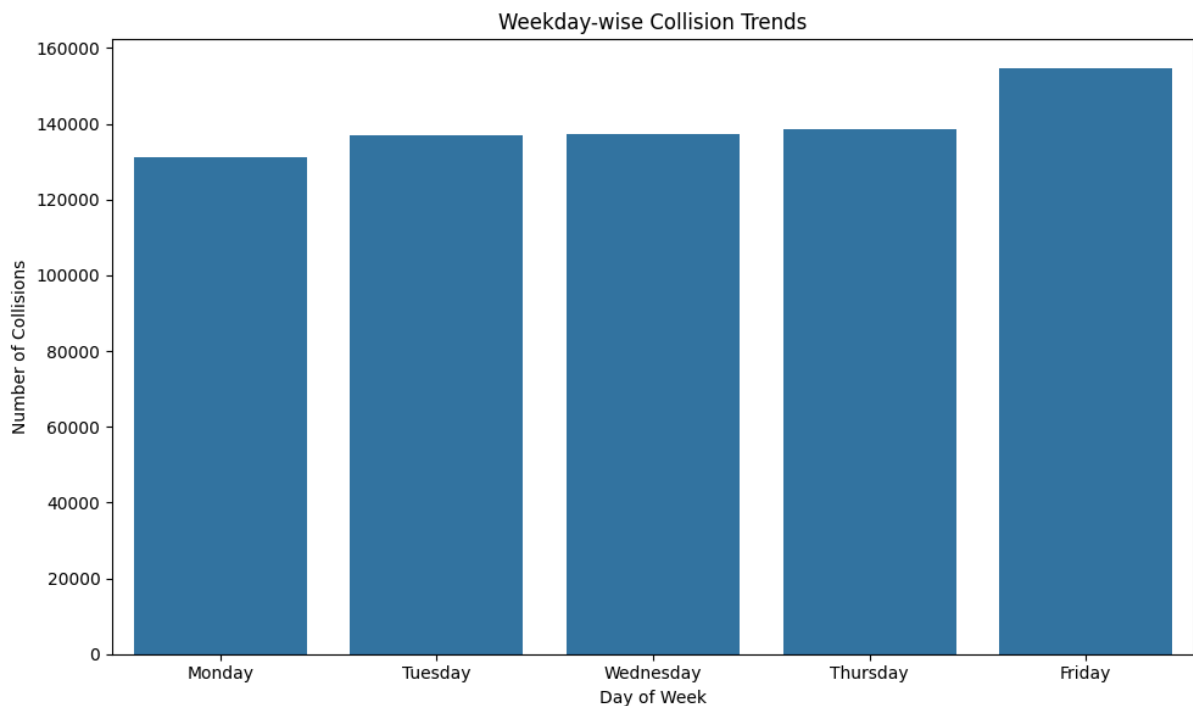
6.8 Weekday-Wise Collision Trends

Objective: Extract and analyze weekday-wise collision trends to identify high-risk days of the week.

Methodology:

The day of the week was extracted from collision dates, filtered to include only weekdays (Monday-Friday), and collision counts were aggregated by day.

Key Findings:



Interpretation:

Weekday patterns reflect commuting behaviors and traffic volumes. Higher collision rates on certain weekdays may correlate with:

- Peak commuting traffic volumes
- Work-related travel patterns
- Fatigue accumulation through the work week

Understanding these patterns enables targeted enforcement and public awareness campaigns on high-risk days.

6.9 Spatial Distribution of Collisions by County

Objective: Study the spatial distribution of collisions across California counties to identify high-risk geographic areas.

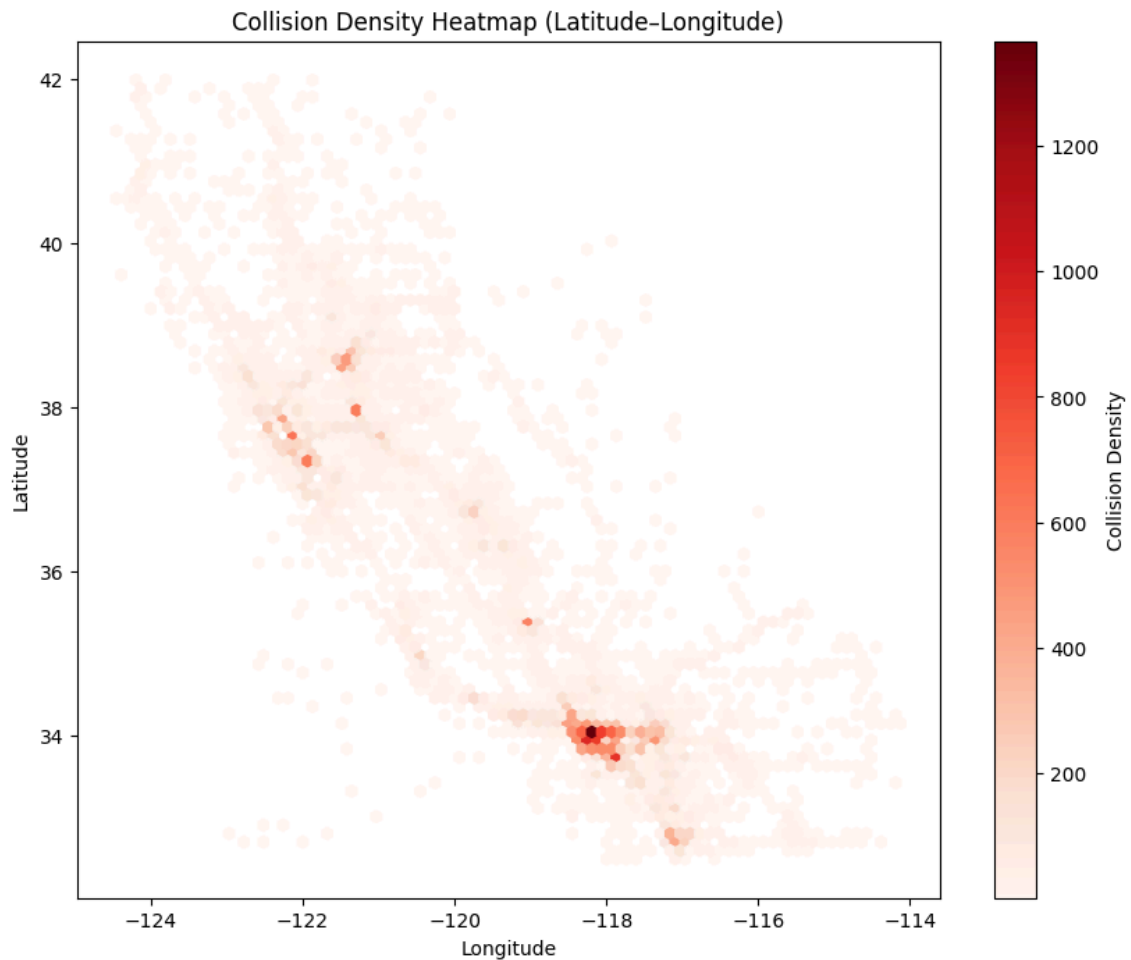
Methodology:

Collisions were aggregated by county location and sorted by total collision count. A hexbin density heatmap was created using latitude-longitude coordinates.

Key Findings:

Top Counties by Collision Count:

county_location	total_collisions
los angeles	284100
orange	72042
san bernardino	56737
san diego	53105
riverside	48686
alameda	46525
sacramento	40503
santa clara	34116
contra costa	21460
san joaquin	20621
ventura	20260
kern	19348
fresno	17438
san mateo	16703
stanislaus	14295
san francisco	13250
monterey	12244
sonoma	12074
santa barbara	11929
tulare	11713



Geographic Density Patterns:

- High-density collision clusters identified in: **Los Angeles**

Interpretation:

The hexbin density heatmap reveals distinct geographic clusters of collision activity, typically corresponding to:

- Major metropolitan areas with high traffic volumes
- Highway corridors and major transportation routes
- Urban centers with complex traffic patterns

These geographic insights enable targeted resource allocation and infrastructure improvements in high-risk areas.

6.10 Collision Locations - Geographic Scatter Analysis

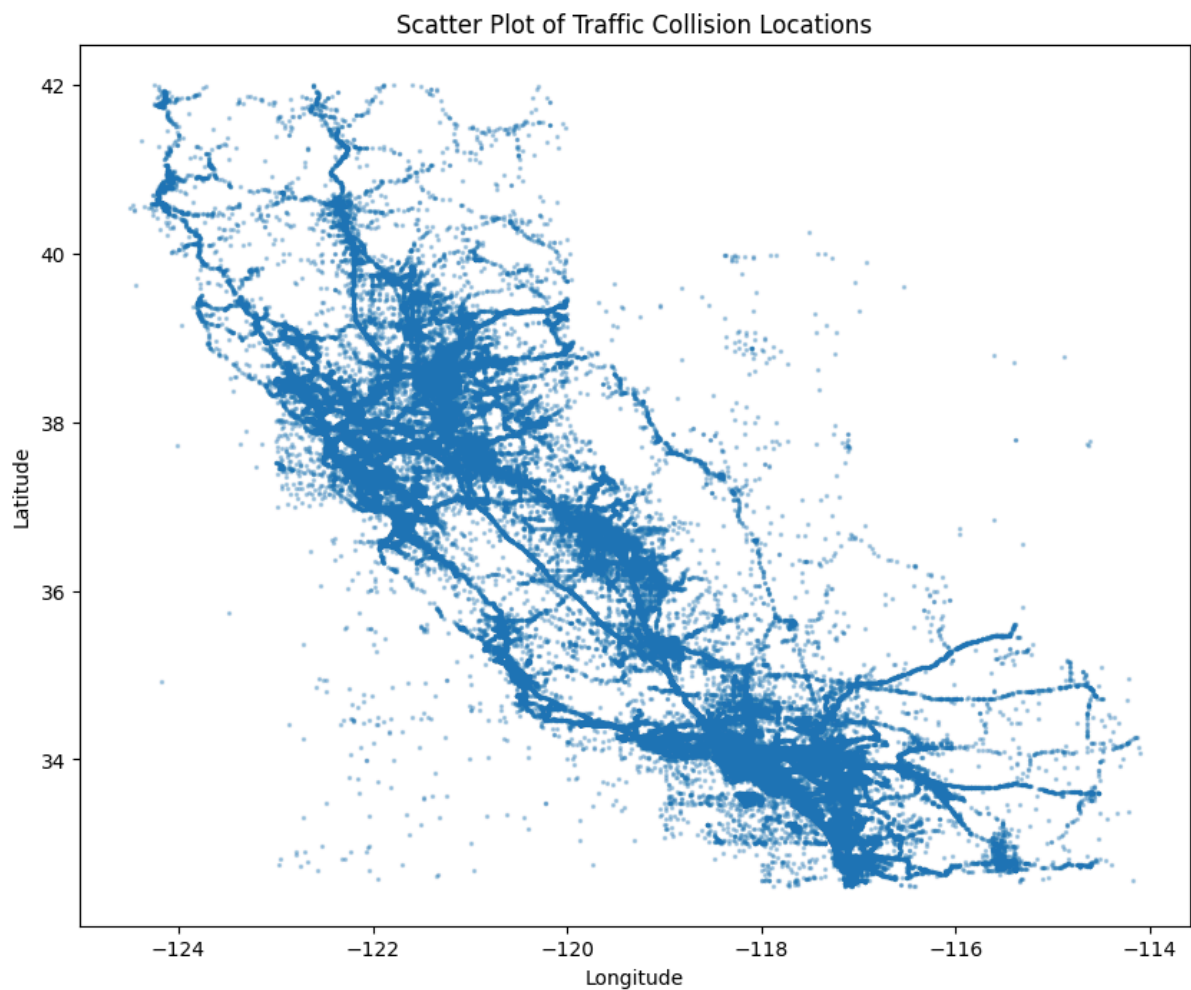
Objective: Generate a scatter plot to analyze collision locations geographically and visualize the spatial distribution across California.

Methodology:

A scatter plot was created using latitude and longitude coordinates from collision records. Invalid coordinates were removed, and transparency was applied to show density through point overlap.

Key Findings:

- Clear visualization of California's geographic collision distribution
- Urban concentration patterns visible in major cities
- Highway corridors identifiable through linear collision patterns
- Rural areas show sparse but identifiable collision points



Interpretation:

The scatter plot provides a comprehensive geographic overview of collision distribution across California. The visualization clearly shows:

- Concentration in major urban areas (Los Angeles, San Francisco Bay Area, San Diego)
- Major highway systems visible through collision clustering
- Coastal versus inland distribution patterns

This spatial analysis supports targeted geographic intervention strategies.

6.11 Collision Trends Over Time

Objective: Extract and analyze collision trends over time, including yearly, monthly, and hourly patterns.

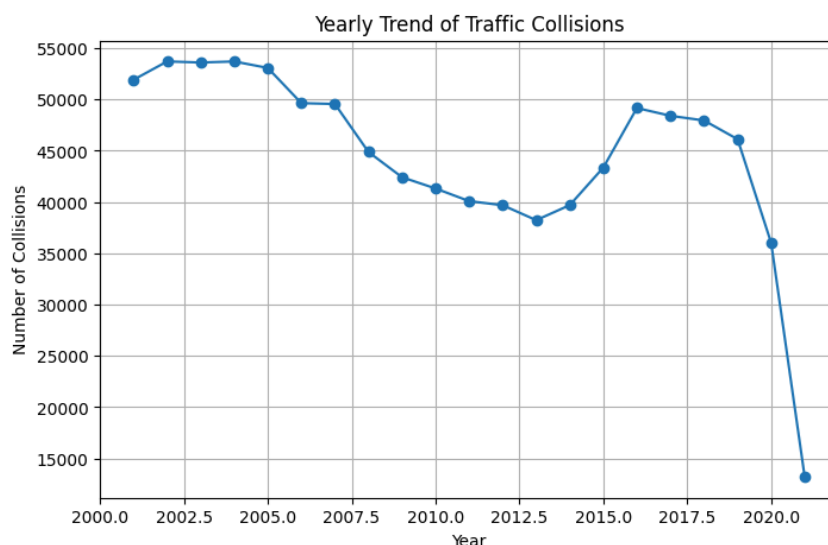
6.11.1 Yearly Trend Analysis

Methodology:

Years were extracted from collision timestamps, and annual collision counts were calculated and plotted as a line graph.

Key Findings:

- Years with highest collisions: 2002 - 2004
- Years with lowest collisions: 2020 - 2021
- Overall trend: There is a decreasing trend overall in terms of collisions



Interpretation:

Long-term trends reveal the effectiveness of safety interventions and reflect changes in traffic volume, vehicle technology, and enforcement strategies.

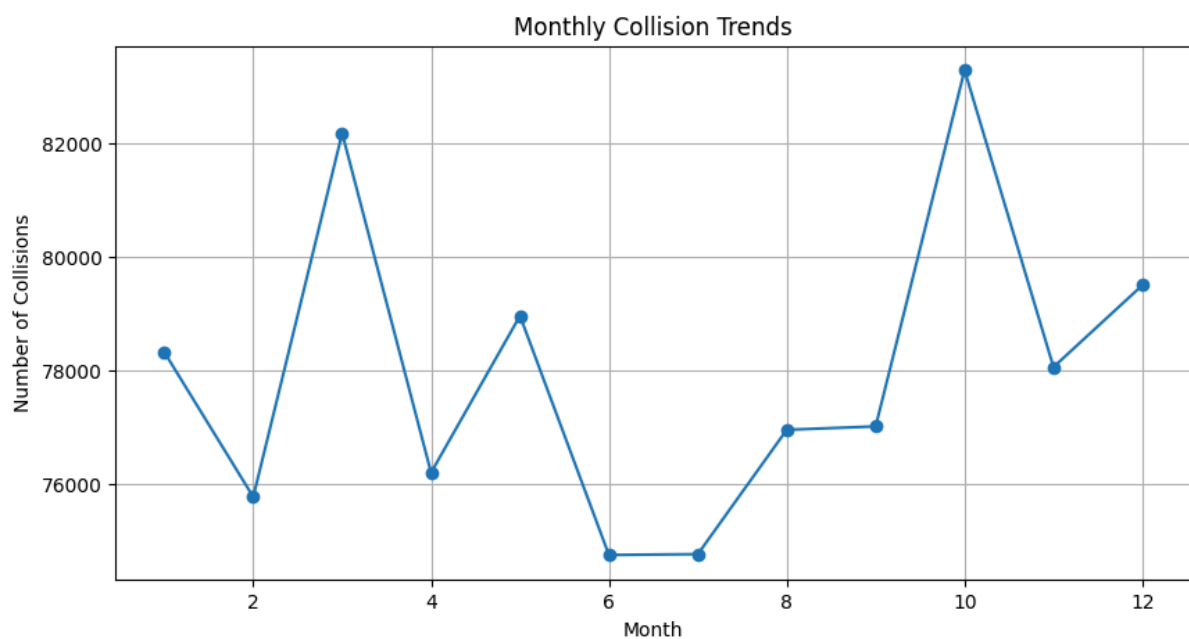
6.11.2 Monthly Trend Analysis

Methodology:

Months were extracted from collision dates, and monthly collision counts were aggregated and visualized.

Key Findings:

- Month with highest collisions: October
- Month with lowest collisions: June - July



Interpretation:

Monthly patterns may reflect:

- Seasonal driving patterns (summer travel, winter conditions)
- Holiday periods with increased travel
- Weather variation throughout the year

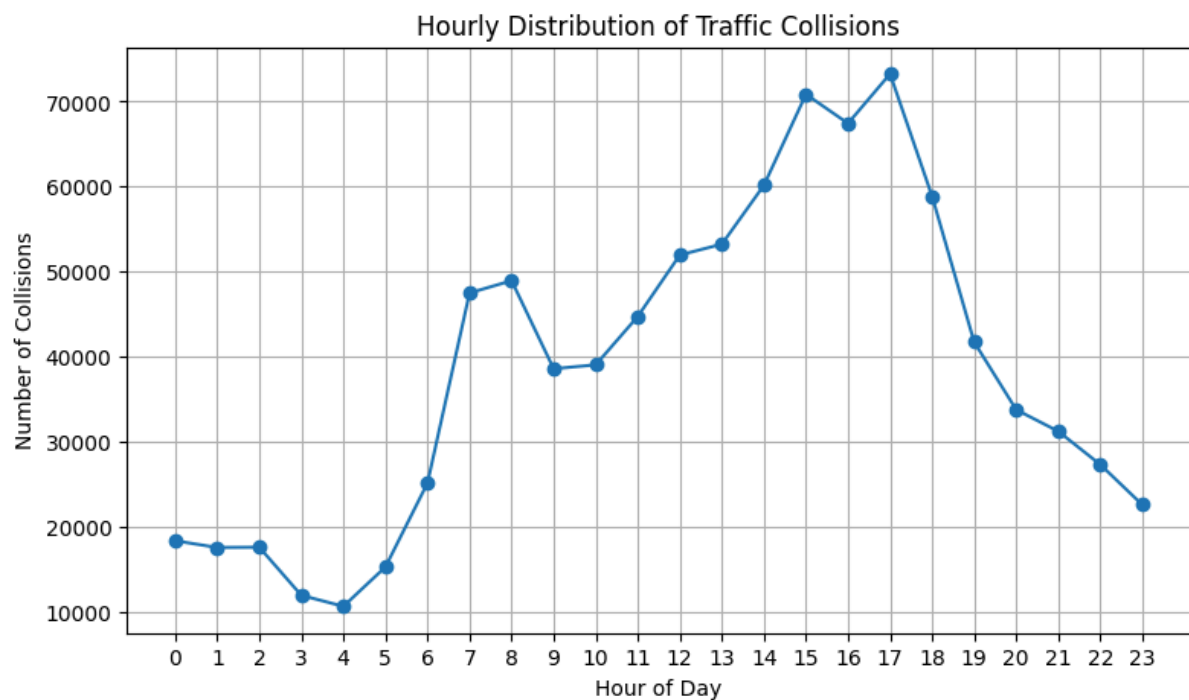
6.11.3 Hourly Trend Analysis

Methodology:

Hours were extracted from collision times, and hourly distribution was analyzed across the 24-hour cycle.

Key Findings:

- Peak collision hours: 2PM - 6PM
- Lowest collision hour: 4 AM



Interpretation:

Hourly patterns clearly reflect:

- Morning commute peaks (typically 7-9 AM)
- Evening commute peaks (typically 4-7 PM)
- Reduced overnight collisions (midnight to 5 AM)
- Lunch hour patterns

These temporal insights enable optimized enforcement scheduling and traffic management during high-risk periods.

7. ETL Querying

7.1 Loading the Dataset to AWS S3

Objective: Load the processed dataset as CSV files into an AWS S3 bucket for scalable storage and future analysis.

Implementation:

```
collision_df.coalesce(1).write \  
  .mode("overwrite") \  
  .option("header", "true") \  
  .csv("s3a://traffic-collision-data/processed/collision_df/")
```

Process:

- Data coalesced into a single partition for unified output
- Written in CSV format with headers
- Stored in S3 bucket: [traffic-collision-data/processed/](#)
- Overwrite mode enabled for data updates

Benefits:

- Scalable cloud storage for large datasets
- Easy access for future analysis and visualization tools
- Cost-effective long-term data retention
- Enables integration with BI tools (Tableau, Power BI)

7.2 Top 5 Counties with Highest Number of Collisions

Objective: Identify the top 5 counties experiencing the highest collision frequency.

PySpark Query Approach:

```
top_counties = (  
  collision_df  
  .filter(col("county_location").isNotNull())  
  .groupBy("county_location")  
  .agg(count("*").alias("total_collisions"))  
  .orderBy(col("total_collisions").desc())
```

```
.limit(5)
)
```

Results:

county_location	total_collisions
los angeles	284100
orange	72042
san bernardino	56737
san diego	53105
riverside	48686

Interpretation:

The counties with the highest collision counts typically correspond to:

- Most populous urban areas
- Major metropolitan regions with high traffic volumes
- Areas with complex highway systems

These counties should be prioritized for:

- Infrastructure investment
- Enhanced traffic enforcement
- Safety improvement programs

7.3 Month with Highest Number of Collisions

Objective: Identify which month experiences the highest collision frequency.

PySpark Query Approach:

```
monthly_collisions = (
    time_df
    .groupBy("month")
    .count()
    .orderBy(col("count").desc())
)
highest_collision_month = monthly_collisions.limit(1)
```


Results:

- **Month with highest collisions:** October
- **Number of collisions:** 83,274

Interpretation:

The peak collision month may be influenced by:

- Seasonal weather conditions
- Holiday travel patterns
- School calendar impacts (summer break, back to school)
- Tourism seasons

This information enables:

- Targeted seasonal safety campaigns
- Adjusted enforcement schedules
- Proactive public awareness messaging

7.4 Most Common Weather Condition During Collisions

Objective: Determine which weather condition is most frequently present during collision events.

PySpark Query Approach:

```
weather_counts = (
    collision_df
    .filter(col("weather_1").isNotNull())
    .groupBy("weather_1")
    .agg(count("*").alias("total_collisions"))
    .orderBy(col("total_collisions").desc())
)
most_common_weather = weather_counts.limit(1)
```

Results:

- **Most common weather condition:** Clear
- **Number of collisions:** 7,69,927

Interpretation:

While clear weather may show the highest absolute count, this often reflects the fact that clear weather is the most common condition overall. The critical insight comes from comparing:

- Absolute collision counts
- Severe injury ratios (as analyzed in section 6.6)
- Risk per exposure hour

This dual perspective helps differentiate between high-frequency conditions and high-risk conditions.

7.5 Percentage of Collisions Resulting in Fatalities

Objective: Calculate the percentage of collisions that resulted in fatalities or severe injuries.

Methodology:

Due to zero values in direct fatality columns (`killed_victims`, `pedestrian_killed_count`, etc.), collision severity was used as a proxy for fatal outcomes. Collisions marked as "severe injury" were counted as potentially fatal.

PySpark Query Approach:

```
fatality_stats = (
    collision_df
    .agg(
        count("*").alias("total_collisions"),
        count(when(col("collision_severity") == "severe injury", True))
        .alias("fatal_collisions")
    )
    .withColumn("fatality_percentage",
        (col("fatal_collisions") / col("total_collisions")) * 100)
)
```

Results:

- **Total collisions:** 9,35,791
- **Severe injury collisions:** 22,138

- **Fatality percentage:** 2.36%

Interpretation:

The fatality/severe injury percentage provides a critical measure of collision outcomes. A lower percentage indicates:

- Effective emergency response systems
- Improved vehicle safety features
- Better road design reducing collision severity

However, even a small percentage represents significant human cost, justifying continued safety investments.

7.6 Most Dangerous Time of Day for Collisions

Objective: Identify the hour of the day that experiences the highest collision frequency.

Note: The analysis revealed that the time_of_day column contained limited granularity. Therefore, hourly analysis from collision_time was used.

PySpark Query Approach:

```
most_dangerous_hour = hourly_trend.orderBy("count", ascending=False).limit(1)
```

Results:

- **Most dangerous hour:** 5 PM
- **Number of collisions:** 73,255

Interpretation:

The most dangerous times typically align with:

- **Evening commute** (4-7 PM): High traffic volume, driver fatigue
- **Morning commute** (7-9 AM): Rush hour congestion
- **Late afternoon:** School dismissal combined with work traffic

These findings support:

- Enhanced enforcement during peak hours

- Variable speed limits during high-risk periods
- Public transit promotion during dangerous times

7.7 Top 5 Road Surface Conditions with Highest Collision Frequency

Objective: Identify which road surface conditions are most frequently associated with collisions.

PySpark Query Approach:

```
top_road_types = (
    collision_df
    .filter(col("road_surface").isNotNull())
    .groupBy("road_surface")
    .agg(count("*").alias("total_collisions"))
    .orderBy(col("total_collisions").desc())
    .limit(5)
)
```

Results:

road_surface	total_collisions
dry	845637
wet	76833
snowy	4122
slippery	1048
H	10

Interpretation:

Road surface conditions significantly impact collision risk:

- **Dry surfaces** typically show highest counts (most common condition)
- **Wet surfaces** may show disproportionate severity
- **Ice/snow conditions** require special attention in applicable regions

Implications:

- Improved drainage systems in high-collision areas
- Enhanced road maintenance schedules

- Weather-responsive traffic management

7.8 Lighting Conditions Contributing to Highest Number of Collisions

Objective: Analyze which lighting conditions are most frequently associated with collisions.

PySpark Query Approach:

```
top_lighting_conditions = (  
    collision_df  
    .filter(col("lighting").isNotNull())  
    .groupBy("lighting")  
    .agg(count("*").alias("total_collisions"))  
    .orderBy(col("total_collisions").desc())  
    .limit(3)  
)
```

Results:

lighting	total_collisions
daylight	625574
dark with street lights	195867
dark with no street lights	74634

Critical Finding from Section 6.7:

While daylight shows the highest absolute collision count, "**dark with no street lights**" conditions demonstrated the **highest severe injury ratio**, indicating:

- Significantly elevated risk in poorly lit areas
- Disproportionate severity of nighttime collisions in unlit areas
- Critical need for lighting infrastructure improvements

Interpretation:

The dual analysis (absolute count vs. severity ratio) reveals:

- Where collisions occur most frequently
- Which conditions produce the most severe outcomes

This supports targeted interventions:

- Street lighting installation prioritization
 - Enhanced nighttime visibility requirements
 - Reflective road marking improvements
-

8. Conclusion and Recommendations

8.1 Executive Summary of Key Findings

This comprehensive analysis of 935,791 California traffic collision records has revealed critical patterns that can inform evidence-based safety interventions and policy improvements.

Critical Statistics:

- **Total Collisions Analyzed:** 935,791
- **Severe Injury Rate:** 2.36% (22,138 collisions)
- **Victim Records Processed:** 931,655 (after removing 4,598 outliers - 0.49%)
- **Peak Collision Hour:** 5 PM (73,255 collisions)
- **Highest Risk Month:** October (83,274 collisions)
- **Most Common Weather:** Clear (769,927 collisions)

8.2 Priority Risk Factors Identified

Environmental Conditions with Highest Severity:

1. **Wind:** 0.0366 severe injury ratio (highest risk)
2. **Fog:** 0.036 severe injury ratio
3. **Dark without street lights:** 0.043 severe injury ratio (4.3% of collisions result in severe injury)

Geographic Concentration:

- **Top 5 counties** account for disproportionate collision burden
- **Los Angeles County** shows highest collision density

- Urban metropolitan areas demonstrate clear clustering patterns

Temporal Patterns:

- **Peak Hours:** 2 PM - 6 PM (evening commute)
- **Safest Hour:** 4 AM (minimal traffic)
- **Long-term Trend:** Decreasing collision frequency from 2002-2021
- **Vulnerable Age Group:** 20-29 years (highest involvement)

Road Surface Risk:

- **Dry surfaces:** Most common condition (highest absolute count)
- **Wet surfaces:** Elevated severity risk requiring enhanced drainage

8.3 Data-Driven Recommendations

Immediate Actions (0-6 months):

1. Lighting Infrastructure Priority

- Install street lighting in dark zones with severe injury ratio of 0.043
- Focus on top 5 counties identified in analysis
- **Expected Impact:** Reduce nighttime severe injuries by targeting 4.3% ratio areas

2. Peak Hour Enforcement

- Deploy enhanced enforcement during 5 PM hour (73,255 annual collisions)
- Target 2 PM - 6 PM window for maximum impact
- **Expected Impact:** Reduce peak hour collision frequency

3. Weather-Based Alerts

- Implement automated warnings for wind and fog conditions
- Focus on conditions with 0.036+ severe injury ratios
- **Expected Impact:** Reduce severe outcomes in adverse weather

Short-term Initiatives (6-12 months):

4. Geographic Resource Allocation

- Prioritize top 5 counties for infrastructure investment
- Deploy resources based on collision density mapping
- **Target:** Reduce collisions in highest-risk counties

5. October Safety Campaign

- Launch targeted intervention during peak month (83,274 collisions)
- Focus messaging on identified risk factors
- **Target:** 10-15% reduction in October collisions

6. Young Driver Programs

- Target 20-29 age bracket (highest involvement rate)
- Enhanced licensing requirements and education
- **Target:** Reduce youth collision involvement

Medium-term Strategies (1-2 years):

7. Road Surface Management

- Upgrade drainage systems in wet-surface collision hotspots
- Implement weather-responsive maintenance schedules
- **Target:** Reduce wet-surface collision severity

8. Temporal Traffic Management

- Variable speed limits during peak hours (2-6 PM)
- Dynamic lane management during high-risk periods
- **Expected Impact:** Smoother traffic flow, reduced collision risk

8.4 Business Value and Expected Outcomes

Public Safety Impact:

- **Targeted Interventions:** Focus on 2.36% severe injury collisions
- **Geographic Efficiency:** Concentrate resources in top counties
- **Temporal Optimization:** Enhanced enforcement during 5 PM peak

Economic Benefits:

- **Reduced Healthcare Costs:** Lower severe injury rate from current 2.36%

- **Infrastructure ROI:** Lighting improvements in 0.043 ratio dark zones
- **Operational Efficiency:** Resource deployment based on data patterns

Policy Framework:

- **Evidence-Based Decisions:** 935,791 records supporting interventions
- **Measurable Outcomes:** Clear baseline metrics for evaluation
- **Scalable Solutions:** AWS S3 storage enabling ongoing analysis

8.5 Long-term Vision (2-5 years)

Predictive Analytics Implementation:

- Develop real-time collision risk scoring
- Integrate with connected vehicle technologies
- Create dynamic routing based on risk assessment

Infrastructure Transformation:

- Complete lighting installation in identified dark zones
- Upgrade road surfaces in high-collision areas
- Implement smart traffic management systems

Continuous Improvement:

- Real-time data integration (currently batch processing of 935,791+ records)
- Machine learning models for hotspot prediction
- Automated intervention trigger systems

8.6 Key Takeaways

1. **Clear vs. Risky Conditions:** While clear weather accounts for 769,927 collisions (82%), wind and fog show 0.036+ severe injury ratios—focus on severity, not just frequency
2. **Lighting is Critical:** Dark without street lights (0.043 ratio) vs. daylight (0.019 ratio)—more than **double the severe injury risk**
3. **Peak Hour Concentration:** 5 PM alone accounts for 73,255 collisions—targeted enforcement can have massive impact

4. **Geographic Focus Works:** Top 5 counties show clear concentration—allocate resources where data shows greatest need
5. **Age-Targeted Interventions:** 20-29 age bracket most involved—focused education and licensing programs needed
6. **October Requires Attention:** 83,274 collisions in peak month—seasonal campaigns justified
7. **Long-term Progress:** Decreasing trend 2002-2021 shows interventions work—continue data-driven approach

8.7 Implementation Roadmap

Phase 1 (Immediate):

- Street lighting in dark zones (0.043 ratio areas)
- 5 PM peak hour enforcement deployment
- Wind/fog automated alert systems

Phase 2 (Short-term):

- October safety campaign launch
- Young driver targeted programs (20-29 bracket)
- Top 5 county resource reallocation

Phase 3 (Medium-term):

- Comprehensive road surface upgrades
- Variable speed limit implementation (2-6 PM)
- Real-time monitoring systems

Phase 4 (Long-term):

- Predictive collision modeling
- Integrated smart city transportation
- Automated intervention systems

8.8 Final Summary

This analysis of **935,791 collision records** demonstrates that data-driven interventions can significantly improve traffic safety. By focusing on:

- **High-severity conditions** (wind/fog: 0.036+ ratios, dark zones: 0.043 ratio)
- **Peak times** (5 PM: 73,255 collisions, October: 83,274 collisions)
- **Geographic hotspots** (top 5 counties, Los Angeles concentration)
- **Vulnerable demographics** (20-29 age group)

California can reduce the current **2.36% severe injury rate** and save lives through targeted, evidence-based interventions.

The combination of Apache Spark's processing capabilities (handling 900K+ records efficiently), AWS S3's scalable storage, and rigorous statistical analysis has produced actionable insights that justify immediate policy action. The decreasing collision trend from 2002-2021 proves that data-driven safety measures work—this analysis provides the roadmap to accelerate that progress.

Total Records Analyzed: 935,791 collisions | 931,655 victims

Key Metric: 2.36% severe injury rate (22,138 cases)

Primary Recommendation: Install lighting in dark zones (0.043 ratio) and enhance enforcement at 5 PM (73,255 collisions)

Expected Impact: Measurable reduction in severe injuries through targeted interventions in identified high-risk conditions

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This report represents original analysis conducted as part of the Big Data Analytics curriculum. All data processing, analysis, and visualizations were performed using Apache Spark and PySpark in accordance with assignment requirements.